

C:\Users\HP\Desktop\Sangar

+

+

Name: Sangam Basnet
Enter a string: abc

Prefixes:

a
ab
abc

Suffixes:

abc
bc
c

Substrings:

a
ab
abc
b
bc
c

Test another string? (y/n): n

×

C:\Users\HP\Desktop\Sangar

Name: Sangam Basnet

Enter a token: int
int -> KEYWORD

Test another token? (y/n): y

Enter a token: def
def -> VALID IDENTIFIER

Test another token? (y/n): y

Enter a token: ldef
ldef -> INVALID IDENTIFIER

Test another token? (y/n): n

C:\Users\HP\Desktop\Sangar

+

+

Name: Sangam Basnet

Enter a string (0s and 1s only): 10010111
10010111 -> ACCEPTED (contains 101)

Test another string? (y/n): y

Enter a string (0s and 1s only): 111001100
111001100 -> REJECTED (doesn't contain 101)

Test another string? (y/n): n

C:\Users\HP\Desktop\Sangar

+

+

Name: Sangam Basnet

Enter a string (0s and 1s only): 10001011
10001011 -> REJECTED (doesn't end with 10)

Test another string? (y/n): y

Enter a string (0s and 1s only): 10101110
10101110 -> ACCEPTED (ends with 10)

Test another string? (y/n): n

C:\Users\HP\Desktop\Sangar

+

+

Name: Sangam Basnet

Enter a string (0s and 1s only): 10011011110
10011011110 -> ACCEPTED (starts with 10)

Test another string? (y/n): y

Enter a string (0s and 1s only): 110101011
110101011 -> REJECTED (doesn't start with 10)

Test another string? (y/n): n

>

+

×

×

×

×

×

×

×

×

×

×

×

×

Name: Sangam Basnet

Enter a string (0s and 1s only): 0011
0011 -> ACCEPTED (0^n 1^n)

Test another string? (y/n): y

Enter a string (0s and 1s only): 011
011 -> REJECTED (not of the form 0^n 1^n)

Test another string? (y/n): n

C:\Users\HP\Desktop\Sangar x + v

Name: Sangam Basnet

Enter a string (0s and 1s only): 1011
1011 -> REJECTED (doesn't end with 10)

Test another string? (y/n): y

Enter a string (0s and 1s only): 110010
110010 -> ACCEPTED (ends with 10)

Test another string? (y/n): n

C:\Users\HP\Desktop\Sangar x + v

Name: Sangam Basnet

Enter a string (0s and 1s only): 11011001
11011001 -> ACCEPTED (contains 101)

Test another string? (y/n): y

Enter a string (0s and 1s only): 11001110
11001110 -> REJECTED (doesn't contain 101)

Test another string? (y/n): n

C:\Users\HP\Desktop\Sangar x + v

Name: Sangam Basnet

Enter a string (0s and 1s only): 100110
100110 -> ACCEPTED (equal number of 0's and 1's)

Test another string? (y/n): y

Enter a string (0s and 1s only): 1110010
1110010 -> REJECTED (unequal number of 0's and 1's)

Test another string? (y/n): n

C:\Users\HP\Desktop\Sangar x + v

Name: Sangam Basnet

Enter a string (0s and 1s only): 0011
0011 -> ACCEPTED ($0^n 1^n$)

Test another string? (y/n): y

Enter a string (0s and 1s only): 011
011 -> REJECTED (not of the form $0^n 1^n$)

Test another string? (y/n): n

>

+

x

C:\Users\HP\Desktop\Sangar

Name: Sangam Basnet

Enter a string (0s and 1s only): 101000
101000 -> ACCEPTED (starts with 10)

Test another string? (y/n): y

Enter a string (0s and 1s only): 1101
1101 -> REJECTED (doesn't start with 10)

Test another string? (y/n): n

>

+

x

C:\Users\HP\Desktop\Sangar

Name: Sangam Basnet

Enter a string (0s, 1s, and 2s only): 001122
001122 -> ACCEPTED ($0^n 1^n 2^n$)

Test another string? (y/n): y

Enter a string (0s, 1s, and 2s only): 01122
01122 -> REJECTED (not of the form $0^n 1^n 2^n$)

Test another string? (y/n): n

Index

S.N.	Title of experiment	Signature
1	Program to find the prefixes, suffixes and substrings from a given input string	
2	Program to simulate a DFA that accepts strings starting with 10 over alphabet $\{0,1\}$	
3	Program to simulate a DFA that accepts strings ending with 10 over alphabet $\{0,1\}$	
4	Program to simulate a DFA that accepts strings containing substring 101 over alphabet $\{0,1\}$	
5	Program to validate valid C identifiers and keywords	
6	Program to simulate a NFA that accepts strings starting with 10 over alphabet $\{0,1\}$	
7	Program to simulate a NFA that accepts strings ending with 10 over alphabet $\{0,1\}$	
8	Program to simulate a NFA that accepts strings containing substring 101 over alphabet $\{0,1\}$	
9	Program to simulate a PDA that accepts strings over $\{0,1\}$ with equal number of 0's and 1's	
10	Program to simulate a PDA that accepts strings over $\{0,1\}$ with equal number of 0's followed by equal number of 1's	
11	Program to simulate a TM that accepts strings over $\{0,1\}$ with equal number of 0's followed by equal number of 1's	
12	Program to simulate a TM that accepts language $L = \{0^n 1^n 2^n \mid n > 0\}$ over alphabet $\{0,1,2\}$	

Tribhuvan University
Institute of Science and Technology



Amrit Campus
Thamel, Kathmandu



B.Sc.CSIT
Lab Report on TOC

Submitted by:

Name: Sangam Basnet

Semester: 4th

Roll No.: 81/081

Symbol No.: 81010244

Submitted to:

Department of CSIT
Amrit Campus, Thamel

Signature